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ELECTRICAL SCIENCE

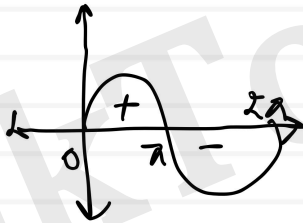
UNIT-2

AC CIRCUITS

AC - current that changes dirⁿ.

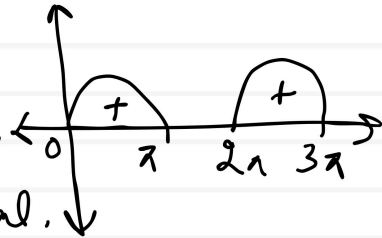
Symmetrical
Waveform :-

when +ve &
-ve part
of waveform
is identical.



Asymmetrical
Waveform

when +ve &
-ve part
of waveform
is unidentical.



INSTANTANEOUS VALUE :-

Value at a particular instant.

$$V_{\text{ins}} = V_0 \sin \omega t$$

AVERAGE VALUE:-

Average Value of all instantaneous values.

symmetrical
waveform

$$V_{avg} = \frac{2}{T} \int_0^{T/2} V d\theta$$

asymmetrical
waveform

$$V_{avg} = \frac{1}{T} \int_0^T V d\theta$$

sinusoidal eqⁿ

$$V_{av} = \int_0^{\pi} \frac{I_0 \sin \theta d\theta}{\pi} = I_0 \frac{(-\cos \theta)}{\pi} \Big|_0^{\pi}$$

$$V_{av} = \frac{2I_0}{\pi} \text{ (half cycle)}$$

$$V_{av} = 0 \text{ (full cycle)}$$

RMS Value :-

$$V_{rms} = \frac{1}{T} \int_0^T V^2 dt$$

$$V_{rms} = \frac{V_0}{\sqrt{2}}$$

Form Factor :- denoted by K_f ; ratio of RMS Value to average value.

$$K_f = \frac{\text{RMS Value}}{\text{Average Value}}$$

Peak Factor :- denoted by K_p , ratio of max^m value to RMS value.

$$K_p = \frac{\text{Max}^m \text{ Value}}{\text{RMS Value}}$$

Active Power - also called real / true power
(P) $P = V_{rms} I_{rms} \cos \phi$

$$P = VI \cos \phi$$

Reactive power -

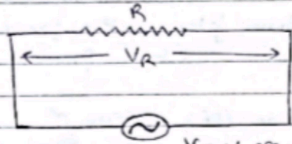
$$= V_{rms} I_{rms} \sin \phi$$

$$= VI \sin \phi$$

Ques

Q An alternating voltage given by $V = 200 \sin \omega t$ and an alternating current is given by $I = 100 \sin (\omega t + 30^\circ)$. Determine power factor, active power, reactive power, apparent power and total power.

SINGLE PHASE purely Resistive circuit



$$V = V_0 \sin \omega t$$

Let $V = V_0 \sin \omega t$ single phase AC supply is given to the purely resistive circuit.

If I is the circuit current then, I can be written as,

$$I = \frac{V}{R} = \frac{V_0 \sin \omega t}{R} \quad \text{--- (2)}$$

If I_0 is maximum value of current then,

$$I_0 = \frac{V_0}{R} \quad [\sin \rightarrow \max \rightarrow \sin]$$

Equation (2) can be written as,

$$I = I_0 \sin \omega t \quad \text{--- (3)}$$

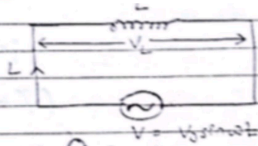
• V & I are in same phase.

$$\therefore \phi = 0 ; \cos \phi = 1$$

• power consumption is maximum.

$$Z (\text{impedance}) = R$$

SINGLE PHASE purely inductive circuit



①
Let $V = V_m \sin \omega t$ single phase AC supply given to the purely inductive circuit.

If I is the circuit current, then V_L can be written as,

$$V_L = L \frac{di}{dt}$$

$$V_m \sin \omega t = L \frac{di}{dt}$$

$$di = \frac{V_m}{L} \sin \omega t dt$$

Now, integrating both sides

$$\int di = \int \frac{V_m}{L} \sin \omega t dt$$

$$\int di = \frac{V_m}{L} \int \sin \omega t dt$$

$$i = \frac{V_m}{\omega L} - \cos \omega t$$

$$i = \frac{V_m}{X_L} \sin \left(\omega t - \frac{\pi}{2} \right) \quad \text{--- (2)}$$

If i_0 is maximum value of current, then equation (2) can be written as,

$$i_0 = \frac{V_m}{X_L} \quad [\sin \omega t \rightarrow 1]$$

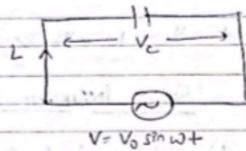
Now, equation (2) can be written as,

$$i = i_0 \sin \left(\omega t - \frac{\pi}{2} \right) \quad \text{--- (3)}$$

- Voltage leads current by $\frac{\pi}{2}$.
- $\cos \phi = 0$
- avg. power = 0
- $Z = X_L = \omega L$

SINGLE PHASE purely capacitive circuit

①
Let $V = V_0 \sin \omega t$ single phase (1- ϕ) AC supply is given to the purely capacitive circuit.



As we know, that

$$q \propto v$$

$$q = CV \quad [C = \text{capacitance}]$$

$$\frac{dq}{dt} = C \frac{dV}{dt} \sin \omega t$$

diff. w.r.t. t

$$i = C \frac{dV}{dt} \sin \omega t$$

$$i = \omega C V_0 \cos \omega t$$

$$i = \frac{V_0}{\frac{1}{\omega C}} \cos \omega t$$

$$i = \frac{V_0}{X_C} \cos \omega t \quad [X_C = \text{capacitive reactance}]$$

$$i = \frac{V_0}{X_C} \sin \left(\omega t + \frac{\pi}{2} \right) \quad \text{--- (2)}$$

If i_0 is maximum value of current then, i can be written as

$$i_0 = \frac{V_0}{X_C} \quad [\sin \rightarrow \max \rightarrow 1]$$

equation (2) can be written as,

$$i = i_0 \sin \left(\omega t + \frac{\pi}{2} \right)$$

• Current leads voltage by $\frac{\pi}{2}$.

• $\cos \phi = 0$ • avg power = 0

$$Z = X_C = \frac{1}{\omega C}$$

Ques. Prove that avg. power in a purely inductive circuit is 0.

$$P_{avg} = \frac{1}{T} \int_0^T P dt$$

$$V = V_0 \sin(\omega t + \frac{\pi}{2})$$

$$i = i_0 \sin \omega t$$

$$P_{avg} = \frac{1}{2\pi} \int_0^{2\pi/\omega} V_0 \sin(\omega t + \frac{\pi}{2}) i_0 \sin \omega t dt$$

$$= \frac{V_0 i_0}{2\pi} \int_0^{2\pi/\omega} \sin \omega t \cos \omega t dt$$

$$= \frac{V_0 i_0 \omega}{4\pi} \int_0^{2\pi/\omega} \sin 2\omega t dt$$

Multiply and divide by 2

$$= \frac{V_0 i_0 \omega}{4\pi} \int_0^{2\pi/\omega} 2 \sin \omega t \cos \omega t dt$$

$$P_{avg} = \frac{V_0 i_0 \omega}{4\pi} \int_0^{2\pi/\omega} \sin 2\omega t dt$$

$$= \frac{V_0 i_0 \omega}{4\pi} \left[-\frac{\cos 2\omega t}{2\omega} \right]_0^{2\pi/\omega}$$

$$= \frac{V_0 i_0 \omega}{4\pi} \left[-\frac{\cos 2\omega \times \frac{2\pi}{\omega} - \cos 0}{2\omega} \right]$$

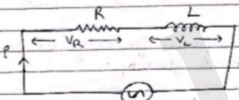
$$= \frac{V_0 i_0 \omega}{4\pi} \left[-\frac{1 - 1}{2\omega} \right]$$

$$= 0 \quad \text{Hence, proved!}$$

Ques. Prove that avg. power in a purely capacitive circuit is 0.

⇒ Series R-L circuit

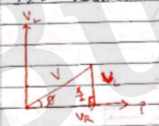
R-L SERIES CIRCUIT



Let $V_R = I R =$ voltage drop across resistance.

$V_L = I X_L =$ voltage drop across inductance

$V = I Z =$ resultant voltage



is value throughout same except the x -axis perpendicular our V_R x -axis pure vol voltage lead I 90° angle.

After V_L ke $\parallel$ the V_R our make resultant V .

$H^2 = R^2 + P^2$
 $V^2 = V_R^2 + V_L^2$

⇒ $Z^2 = R^2 + X_L^2$

$Z = R^2 + X_L^2$

$Z = \sqrt{R^2 + X_L^2}$

The value of power factor,

$$\cos \phi = \frac{V_R}{V}$$

$$= \frac{I R}{I Z}$$

$$\boxed{\cos \phi = \frac{R}{Z}}$$

The expression for voltage and current for R-L series circuit is given by:

$$V = V_0 \sin \omega t$$

$$I = I_0 \sin (\omega t - \phi)$$

$$I = I_0 \sin \omega t$$

$$V = V_0 \sin (\omega t + \phi)$$

Series R-C Circuit

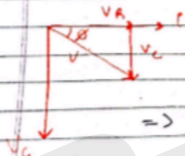
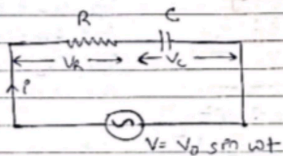
R-C SERIES CIRCUIT

Let,

$$V_R = IR$$

$$V_C = I \times C$$

$$V = IZ$$



$$V^2 = V_R^2 + V_C^2$$

$$V = V_0 \sin \omega t$$

$$\Rightarrow I^2 Z^2 = I^2 R^2 + I^2 C^2$$

$$Z^2 = R^2 + X_C^2$$

$$Z = \sqrt{R^2 + X_C^2}$$

$$X_C = \frac{1}{\omega C}$$

$$\text{The value of power factor (P.F.)} = \cos \phi = \frac{V_R}{V} = \frac{IR}{IZ}$$

$$PF = \cos \phi = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + X_C^2}}$$

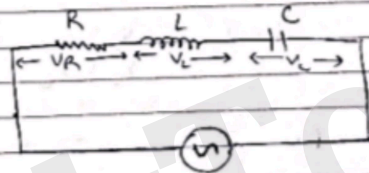
The expression for voltage and current is given by
 $V = V_0 \sin \omega t$

$$I = I_0 \sin (\omega t + \phi)$$

Series R-L-C Circuit:-

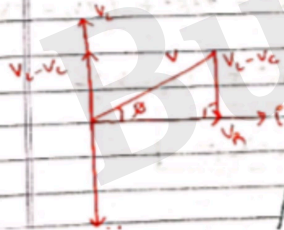
R-L-C SERIES CIRCUIT

- 1) $V_L > V_C$
- 2) $V_C > V_L$
- 3) $V_L = V_C$



Case I: $V_L > V_C$
or $X_L > X_C$

$$[IX_L > IX_C]$$



$$V^2 = V_R^2 + (V_L - V_C)^2$$

$$I^2 Z^2 = I^2 R^2 + (IX_L - IX_C)^2$$

$$Z^2 = R^2 + (X_L - X_C)^2$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\begin{aligned} \text{Power factor, } \cos \phi &= \frac{V_R}{V} \\ &= \frac{IR}{IZ} = \frac{R}{Z} = \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} \\ &= \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}} \end{aligned}$$

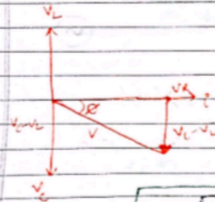
* here voltage will lead by some angle ϕ .

* The nature of power factor is decided from the nature of current.

In this case, Voltage is leading; Current is lagging and Power factor is also lagging.

Case II: $V_C > V_L$

$$X_C > X_L \quad [IX_C > IX_L]$$



$$V^2 = V_R^2 + (V_C - V_L)^2$$

$$I^2 Z^2 = I^2 R^2 + (IX_C - IX_L)^2$$

$$Z^2 = R^2 + (X_C - X_L)^2$$

$$Z = \sqrt{R^2 + (X_C - X_L)^2}$$

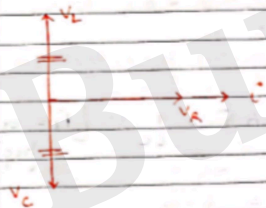
$$\text{Power factor } \cos \phi = \frac{V_R}{V}$$

$$= \frac{IR}{IZ}$$

$$= \frac{R}{Z}$$

$$= \frac{R}{\sqrt{R^2 + (X_L - X_C)^2}}$$

Case III: $V_L = V_C$ (Purely resistive)



$$V^2 = V_R^2 + (V_L - V_C)^2$$

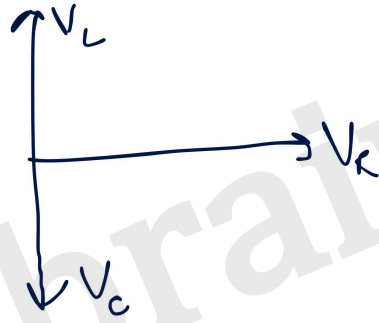
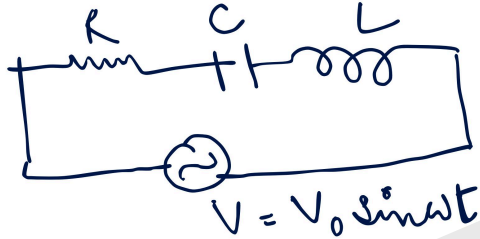
$$V^2 = V_R^2$$

$$I^2 Z^2 = I^2 R^2$$

$$Z = R$$

It means, in this case circuit behaves as purely resistive circuit.

Series Resonance:-



- Resonance is a condition when ind. & cap. resistance balance out one another & hence max^m current flows through the circuit.
- series resonant frequency - value of frequency at which resonance takes place.
- also called acceptor circuit

current $\rightarrow \text{max}^m$

Impedance $\rightarrow \text{min}^m$

for series resonance, voltage across inductance is equal to voltage across capacitance.

$$V_L = V_C$$

$$\Rightarrow I X_L = I X_C$$

$$\Rightarrow X_L = X_C \quad [\text{inductive reactance} = \text{capacitive reactance}]$$

$$\Rightarrow \omega L = \frac{1}{\omega C}$$

$$\Rightarrow \omega^2 = \frac{1}{LC}$$

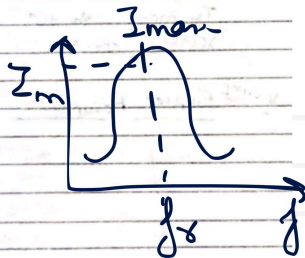
$$= (2\pi f_r)^2 = \frac{1}{LC} \quad [f_r = \text{resonance frequency}]$$

$$= 4\pi^2 f_r^2 = \frac{1}{LC}$$

$$f_r^2 = \frac{1}{4\pi^2 LC}$$

$$f_r = \frac{1}{2\pi \sqrt{LC}}$$

$$f_r = \frac{1}{2\pi \sqrt{LC}}$$



$\Rightarrow$ the net reactance is 0.

$$X_L - X_C = 0$$

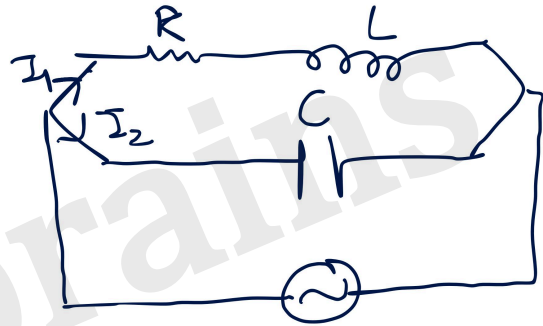
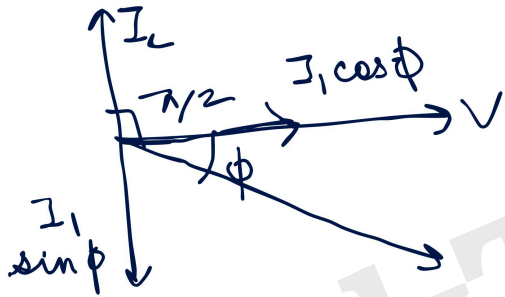
$$\Rightarrow Z = R \quad \Rightarrow I = \frac{V}{Z}$$

$$\Rightarrow \cos \phi = \frac{R}{Z}$$

$\therefore$ It is purely resistive circuit.

$\Rightarrow$ Quality factor - It is defined as ratio of voltage across inductor/capacitor to the total voltage. $QF = \frac{1}{R} \sqrt{\frac{L}{C}}$

PARALLEL RESONANCE



- reactive component is zero.
- In parallel resonance, net current is less as inductor & capacitor current cancel out each other.
- also called rejecting circuit

current $\rightarrow \min^m$
Impedance $\rightarrow \max^m$

Q A coil has an inductance of 0.05H and a resistance of 10Ω . It is connected to a sinusoidal 200V , 50Hz supply. Calculate the impedance, current, power consumed and power factor.

Q

A coil having resistance $12\ \Omega$, inductance 0.05 H , a non inductive resistance of $20\ \Omega$ and loss free $40\ \mu\text{F}$ capacitor are connected in series across 240 V , 50 Hz AC supply.

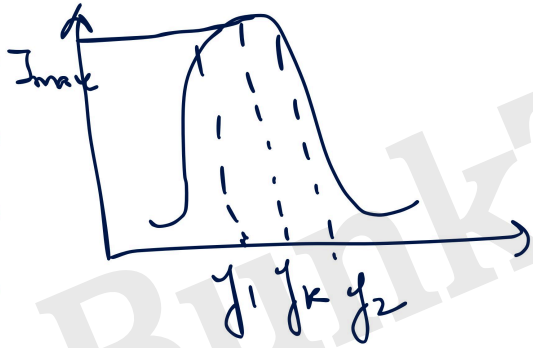
Determine 1) value of total impedance

2) circuit current

3) power factor

4) power delivered across load

BANDWIDTH - range of frequency at which the current is equal to 70.7% of its maximum value.

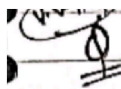


$$BW = f_2 - f_1 \quad f_k = \sqrt{f_1 f_2}$$

$$QF = \frac{f_k}{BW} \rightarrow \text{resonant frequency}$$

Q A single phase load of 30 kW at 0.6 power factor lagging is fed from 200 V a.c. supply. Calculate the kVA and kVAR of the load.

Q Calculate the resonance frequency, bandwidth and lower and upper half power frequency when the value of coil resistance is 10.05Ω and inductance $400 \mu H$ is connected with a series capacitor $0.396 \mu F$.

 A coil having a resistance of $10\ \Omega$ and inductance of 0.05 H is connected in series with a capacitor of $100\ \mu\text{F}$. The whole circuit has been connected to 200 V , 50 Hz AC supply. Calculate

- 1) circuit impedance
- 2) circuit current
- 3) power factor
- 4) active power, reactive power and apparent power.

 A 250 (rms) , 50 Hz voltage is applied across a circuit consisting of a pure (non-inductive) resistance of $20\ \Omega$. Determine

- i) current flowing through circuit
- ii) power absorbed by circuit

Give the expressions for the voltage and current.

SINGLE PHASE AC CIRCUIT:-

Advantages of 3 ϕ AC ckt over 1 ϕ

- (1) The power deliver in single phase is pulsating in nature, but power deliver in 3- ϕ is constant.
- (2) The output of 3- ϕ power is 1.5 times greater than single phase for same size of machine.
- (3) The material required in three phase is $\frac{3}{4}$ th less than single phase.
- (4) The value of power factor and efficiency is greater than 1- ϕ .

$$\uparrow I \propto \frac{1}{\cos \phi} \downarrow \quad \rightarrow I \uparrow \rightarrow \text{loss} \uparrow$$

- (5) The 3- ϕ machine is self-starting, except synchronous motor but 1- ϕ motor is not self-starting.

Balance supply - when all 3 phases are equal in magnitude.

Phase sequence - when phases reach their max^m value one after another.

3 ϕ AC connection —
 3 ϕ star connection
 3 ϕ delta connection.

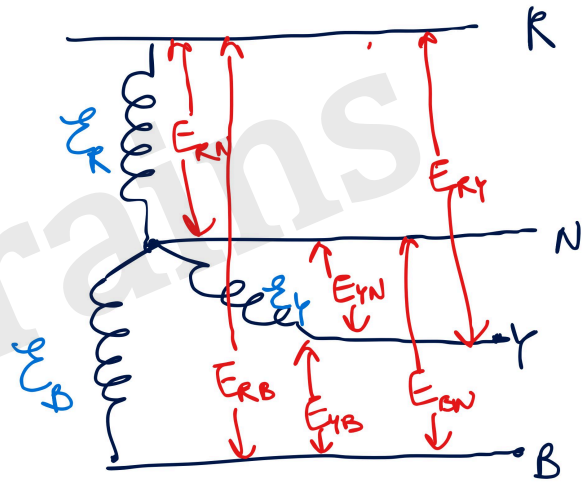
Phase Voltage

The voltage across winding or the voltage b/w any 1- ϕ and neutral is known as Phase Voltage.
It is represented by E_R , E_Y and E_B or E_{RN} , E_{YN} , E_{BN} and line voltage is represented by E_{RY} , E_{YB} and E_{BR} .

for balanced supply

$$E_{RY} = E_{YB} = E_{BR} = E_s$$

$$E_R = E_Y = E_B = E_{RN} = E_{YN} = E_{BN} = E_p$$



Line Voltage - voltage b/w any two phase is known as line voltage.

Ex :- E_{RY}, E_{BR}

Ques Derive the relation b/w phase & line voltage for 3- ϕ star connection

OR

Prove that line voltage is equal to $\sqrt{3}$ times phase voltage.

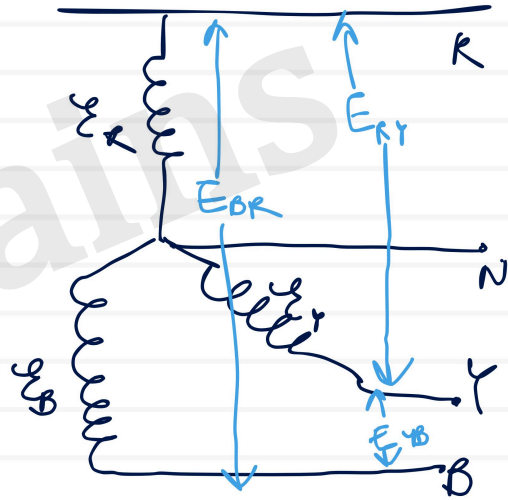
Here E_R, E_Y, E_B are phase voltage
 ξ

E_{RY}, E_{BR}, E_{YB} are line voltage.

For balance supply;

$$E_R = E_Y = E_B = E_p$$

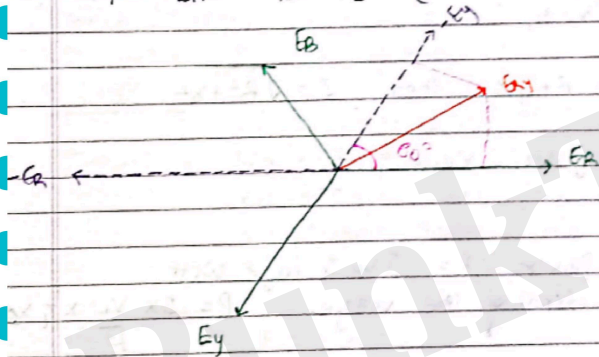
$$E_{RY} = E_{YB} = E_{BR} = E_L$$



Simplify for current

$$I_A = I_Y = I_B = I_P \quad (\text{phase current})$$

$$I_{AY} = I_{BR} = I_{GB} = I_L \quad (\text{line current})$$



$$E_{RY} = \sqrt{E_R^2 + E_Y^2 + 2E_R E_Y \cos 60^\circ}$$

$$E_{RY} = \sqrt{E_R^2 + E_Y^2 + 2E_R E_Y \times \frac{1}{2}}$$

$$E_{RY} = \sqrt{E_R^2 + E_Y^2 + E_R E_Y}$$

$$E_{RY} = \sqrt{E_p^2 + E_p^2 + E_p^2}$$

$$E_L = \sqrt{3} E_p \quad \begin{matrix} \text{in terms of} \\ \text{V (vol)} \end{matrix} \quad V_L = \sqrt{3} V_p \quad \begin{matrix} \text{in terms of} \\ \text{voltage} \end{matrix}$$

Similarly, $E_{YB} = \sqrt{3} E_p$ and,

$$E_{BR} = \sqrt{3} E_p$$

Therefore, line voltage $V_L = \sqrt{3} V_p$.

$$V_L = \sqrt{3} V_p$$

$$\text{active power (P)} = 3 V_p I_p \cos \phi \quad \text{or} \quad \sqrt{3} V_L I_L \cos \phi$$

($\sin \phi$ for reactive power)

Q A 3- ϕ balance star-connected load $(8+j6) \Omega$ per phase is connected to 400V, 50 Hz AC supply. Determine
i) phase current and line current (I_p) (I_L)

ii) power factor

iii) Active power (P)

iv) Reactive power (Q)

v) Apparent power (S)

$\star$ conn.

$$V_L = \sqrt{3} V_p$$

$$I_L = I_p$$

Δ conn.

$$V_L = V_p$$

$$I_L = \sqrt{3} I_p$$

Q A 3- ϕ balanced delta connected load ($16 + j12 \Omega$) per phase is connected to 4150 V, 50 Hz AC supply. Determine i) phase voltage and line voltage
(V_p) (V_L)

ii) Phase current and line current
(I_p) (I_L)

iii) Active Power and Reactive Power
(P) (Q)

iv) Total Volt Amperes (Total Power)
(S)

All the Best for your Exams!



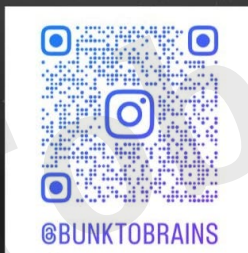
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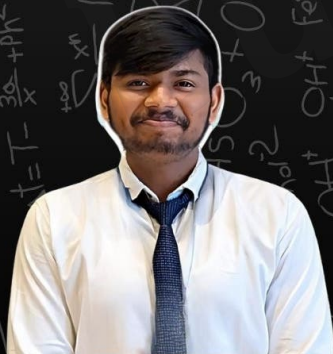


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